SNUC HACKS '26

LOGICASH (LOGIC + CASH)

TRACK : 3 (FINTECH)

TEAM NAME: CLAUDERS
TEAM LEADER: AATISH S
PHONE NUMBER: 7338878319
ROOM NO: 127 (1ST FLOOR)

PROBLEM STATEMENT

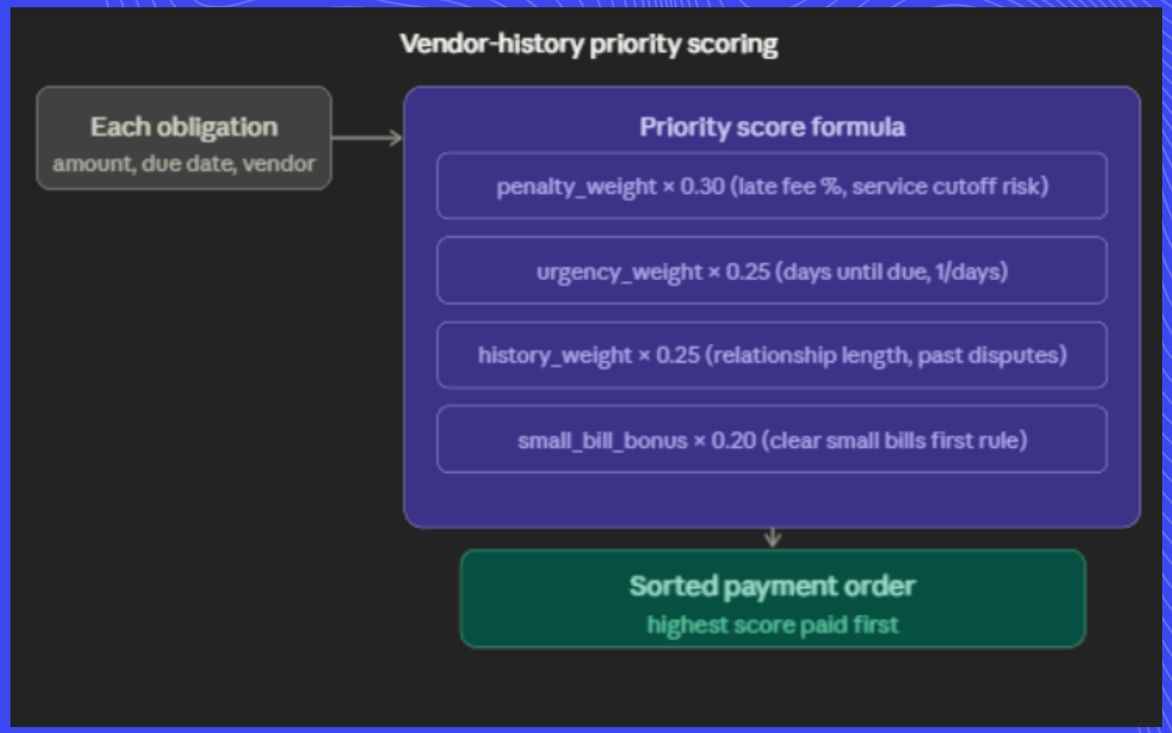
Problem Area	Current Gap	Impact on Businesses	Our Resolution
Financial Visibility	Data scattered across receipts, invoices, banks	No unified view of obligations vs cash	Aggregates & structures all financial inputs into one model
Payment Prioritization	No system to decide what to pay first	Random / suboptimal payment decisions	Deterministic engine ranks obligations intelligently
Trade-off Analysis	Cannot evaluate penalties vs relationships vs survival	Poor decision-making under pressure	Multi-factor scoring (penalty, urgency, vendor trust, flexibility)
Lack of Decision Support	Tools only track data, don't guide actions	Reactive, intuition-based decisions	Converts financial data → clear, actionable decisions
Vendor Communication	Manual, delayed, poorly framed communication	Relationship damage, negotiation failures	Auto-generates context-aware negotiation emails
Surplus Utilization	No guidance when extra cash exists	Missed growth opportunities	Suggests business opportunities with ROI insights

SNUC HACKS '26

INNOVATION AND UNIQUENESS

A semi-autonomous system that converts financial data into decisions + actions:

- Unified Financial Model
  - Aggregates receipts, invoices, and bank data into structured obligations
- Deterministic Decision Engine
  - Computes runway & prioritizes payments using urgency, penalties, and flexibility
- Scenario Engine
  - Gives user choices on which factor they prioritise (Eg: Relationship, Penalty, etc)
- User Preference based Scoring
  - Calculate scores based on the Preferences chosen by the user
- Action Layer
  - Produces payment plans + Automated Vendor negotiation emails
- Surplus Intelligence
  - Suggests business opportunities when excess cash exists



SNUC HACKS '26

TECHNICAL ARCHITECTURE

LAYER 1 : Authentication

Google Oauth 2.0

LAYER 2 : Input Layer

- Photo Upload (Tesseract)
- PDF Invoice (PyMuPDF)
- Bank Statement (CSV Parse)
- Manual Upload (Salaries, etc)

LAYER 3 : Storage

Vendor Profile & History (Supabase)

LAYER 4 : Logical Reasoning (Purely using python)

- Runway Calculation (Days to Bankrupcy)
- Priority Scorer (Vendor reputation)
- Conflict Detector (Unable to pay dues)
- Conflict Detector (Unable to pay dues)

LAYER 5 : Output Layer

- Email Drafter (Tone aware along with justifications)
- Oppurtunity Finder (Incase of surplus funds)
- Exportable Output (in PDF format)

SNUC HACKS '26

FEASIBILITY AND VIABILITY

- Feasibility & Viability
  - Technical Feasibility
    - Built using proven tools (Python, Tesseract, React, Gemini API)
    - Core decision logic is deterministic and testable
    - Modular design enables parallel development and quick integration
  - Implementation Viability
    - Core system (input + prioritization)
    - Advanced features (OCR, scenarios, emails) added incrementally
    - Easy deployment via StreamLit
- Robustness
  - Hybrid approach (rules + OCR + AI) ensures:
  - Flexibility for real-world messy data
  - Reliability in financial decision-making
- Use Cases
  - Cash flow prioritization under constraints
  - Financial conflict resolution
  - Vendor communication automation
  - Surplus-driven business planning
- Target Users
  - Small & medium businesses (SMBs)
  - Freelancers & startups without CFO support
  - Businesses with recurring operational expenses
- Usage
  - Used daily/weekly for expense tracking and decisions
  - Critical during cash shortages
  - Monthly for financial planning

SNUC HACKS '26

IMPACT AND BENEFITS

- Intelligent Financial Decision-Making
  - Moves beyond tracking → actively decides what to pay, when, and why
  - Eliminates guesswork during cash flow constraints
- Improved Cash Flow Survival
  - Runway-based prioritization ensures business longevity
  - Prevents critical failures (missed salaries, rent, taxes)
- Optimized Payment Strategy
  - Balances penalties, urgency, and vendor relationships
  - Generates multiple scenarios for smarter trade-offs
- Stronger Vendor Relationships
  - Prioritizes high-impact relationships
  - Auto-generates professional communication for delays
- Time & Cognitive Load Reduction
  - Automates complex financial reasoning
  - Saves hours of manual decision-making
- Transparency & Trust
  - Every decision is explainable and human-readable
  - No “black-box” financial logic
- Growth Enablement
  - Detects surplus and suggests business opportunities
  - Shifts mindset from reactive → strategic

RESEARCH AND REFERENCES

- Background Research
  - Small businesses rely on bank balance visibility instead of forward planning
  - High failure rate due to poor cash flow management and delayed decisions
  - Financial decisions involve trade-offs between penalties, relationships, and survival
  - Existing tools lack real-time decision support and explainability

◆ Comparative Analysis

Tool	What it Does	Gap
QuickBooks	Expense tracking, reports	No decision-making
Zoho Books	Accounting automation	No prioritization logic
Excel/manual	Flexible tracking	No intelligence or automation

- Uniqueness
  - Deterministic payment prioritization engine
  - Scenario-based decision making
  - Explainable financial reasoning
  - Actionable outputs (emails, plans)

SNUC HACKS '26

SNUC HACKS '26

THANK YOU!

TEAM NAME: CLAUDERS
TEAM LEADER: AATISH S
PHONE NUMBER: 7338878319
ROOM NO: 127 (1ST FLOOR)